

Submit the following five exercises. You are expected to provide full solutions in a linear and coherent fashion. You are encouraged to work in groups, but your work must be your own and must be self-contained.

Exercise 1.

Find the domain of the function

$$f(x) = \frac{\sqrt{x-1}}{x^2 - x - 2}.$$

Exercise 2.

Find all real roots (zeros) of

$$g(x) = \begin{cases} x^2 + 1 & \text{if } x < 0, \\ \cos(x) & \text{if } 0 < x < 2, \\ -2x + 6 & \text{if } 2 < x. \end{cases}$$

Exercise 3.

For $0 \leq \theta < 2\pi$, find all real roots (zeros) of

$$h(x) = 6 \sin(3\theta) - 2\pi.$$

Exercise 4.

Find all real roots (zeros) of

$$i(x) = x^4 - 4x^2.$$

Exercise 5.

Find all real values of c such that

$$j(x) = \frac{x-1}{x^2 + 2cx + 4}$$

has domain all real numbers $\mathbb{R}$.

Additional recommended exercises (do not submit).

Section 1.1:

39, 40, 41-48, 49-52, 53-58, 59, 63, 64, 65-68, 71, 72, 74, 75, 76, 77.

Section 1.2:

1-4, 9-20, 23, 24, 25, 26, 27, 28, 29, 33, 34-41, 42, 43, 45, 46, 50, 51.

Section 1.3:

1-12, 26, 27-34, 36.

Section 1.4:

1, 2, 3, 4, 7, 9-14, 15, 16, 31.

Section 1.5:

2, 3, 4, 6, 21, 22.